International Journal of Geo-Information

MDPI

Article

Spatially Explicit Assessment of Climate Gentrification in Miami-Dade County: Elevation, Sea Level Rise, and Demographic Displacement

Patrick Davenport * and Levente Juhász

Geospatial Analytics, Technology and Open Research (GATOR) Laboratory, Fort Lauderdale Research and Education Center, University of Florida, Davie, FL 33314, USA; levente.juhasz@ufl.edu

* Correspondence: davenport.p@ufl.edu

Abstract

This study examines the spatial dynamics of climate-driven gentrification across 646 census tracts in Miami-Dade County, Florida, from 2010 to 2023, testing whether sea level rise exposure and elevation are associated with income growth, home value appreciation, and Black non-Hispanic demographic displacement, using American Community Survey data and NOAA 2.1 m inundation. County-wide Ordinary Least Squares (OLS) models indicate that elevation is not associated with income change, while elevation is significantly associated with Black non-Hispanic population decline in historically Black tracts (6.6 percentage points per meter where the 2010 share was at least 30%). Geographically Weighted Regression (GWR) indicates that this elevation–displacement relationship is spatially heterogeneous, concentrated in northern Miami-Dade and improves on OLS (mean local $R^2 = 0.57$ vs. 0.35). Among high-elevation tracts (mean elevation > 2.1 m), GWR ($R^2 = 0.49$ vs. 0.33) shows that baseline socioeconomic vulnerability is associated with gentrification intensity, while elevation retains a positive association. These findings extend climate gentrification theory, which centers on property value appreciation, by suggesting that demographic displacement was a more sensitive indicator than income or home value change in Miami-Dade. Nesting global county-wide and focused analyses suggests that climate gentrification may not be detectable at a single analytical scale, and income-based metrics may underestimate the phenomenon.

Keywords: climate gentrification; sea level rise; Geographically Weighted Regression; Miami-Dade County; demographic displacement; elevation; flood risk; environmental justice

Check for updates

Academic Editor: Godwin Yeboah

Received: 26 August 2026
Revised: 23 October 2026
Accepted: 24 October 2026
Published: 28 October 2026

Copyright: © 2026 by the authors. Published by MDPI on behalf of the International Society for Photogrammetry and Remote Sensing. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

As sea levels rise at accelerating rates [1], coastal cities face not only the direct physical consequences of inundation and flooding but socioeconomic ones as well. This acceleration is acutely evident in Miami, where local tide gauge data from the Virginia Key station show the rate of sea level rise (SLR) increased from 6.36 ± 0.35 mm per year (measured over 1993–2022) to 9.36 ± 1.05 mm per year during the 21st century (2003–2022) [2]. How urban land is valued and occupied is being reshaped by this localized climate risk, and few places in the continental United States illustrate this dynamic more clearly than Miami-Dade County (MDC). Situated along the southeastern coast of Florida, MDC ranks among the most vulnerable urban jurisdictions in the country due to its low-lying topography, porous

limestone geology and decades of coastal urban development [3]. First formalized by Keenan et al. [3], climate gentrification theory posits that climate-driven shifts in flood and inundation risk differentially affect property values, attracting higher-income residents and investment to climate-resilient areas while displacing lower-income populations who can no longer afford to remain.

MDC's low-lying topography and large urban population where residents already face the impacts of SLR make it a particularly compelling case study [4]. The left panel in Figure 1 shows that much of the county would be inundated or be sitting below the high water line under 2.1 m of SLR. Unlike many built environments, where wealth has historically concentrated at higher elevations [5], low-lying coastal areas in MDC attracted affluent development throughout much of the 20th century. Simultaneously, segregation-era planning policies displaced marginalized communities inland to areas that were considered less desirable precisely because they were further from the waterfront [6]. This historical settlement pattern has a spatial consequence that is only now becoming apparent: those inland areas, situated along Miami's Atlantic Coastal Ridge (a limestone formation running through the interior of the county at higher elevations) (Figure 2), have become the most climate-resilient parcels of real estate in the region. This topographic advantage has recently catalyzed intense speculative real estate investment, driving rapid physical transformations in historically marginalized neighborhoods like Overtown (Figure 3). Apart from Overtown, Liberty City and Little Haiti are two other historically Black neighborhoods that also sit at the center of this dynamic.

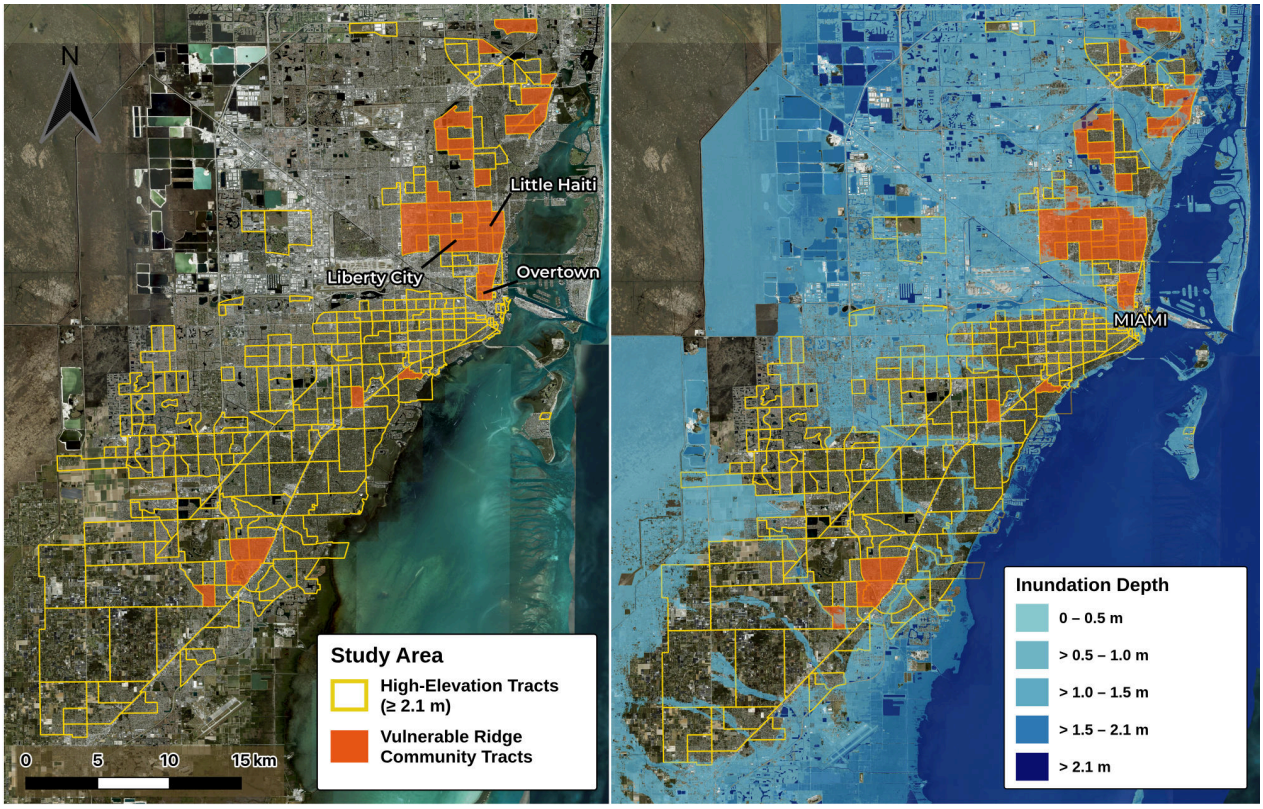


Figure 1. (**Left**) panel: Census tracts at or above 2.1 m (7 ft) NAVD88 elevation (n = 277, yellow outline). (**Right**) panel: Miami-Dade County under a 2.1 m (7 ft) SLR inundation scenario (NOAA depth data [7]), illustrating projected flood extent and depth (see legend) across the study area. Vulnerable Ridge Community (VRC) tracts (n = 42, orange fill) additionally had 2010 median household income below USD 40,000 and Black non-Hispanic population share above 30%, combining flood resilience with historical vulnerability as potential climate gentrification hotspots. Focus neighborhoods (callouts) are Overtown, Little Haiti and Liberty City.

Overtown was established in the early 20th century as a segregated Black community and served as the cultural heart of Black Miami [8]. Highway construction and urban renewal projects in the 1950s and 1960s dismantled much of the neighborhood's built fabric and dispersed its population [6]. Liberty City emerged in the late 1930s as one of Miami's first large-scale public housing communities, which was developed during an era when Black residents were barred from most other parts of the city; over the following decades, it became a major center of cultural, social, and political life for Black Miami even as redlining and disinvestment constrained local economic opportunities. Little Haiti, by contrast, formed more recently, emerging in the 1980s as a key center of Miami's Haitian diaspora that was shaped by successive waves of migration, community institutions and Haitian-owned businesses. Together, these three neighborhoods represent a historical marginalization and present-day climate advantage: each sits on elevated ground along the Atlantic Coastal Ridge (ACR) (Figure 2), making them among the most flood-resilient residential areas in an otherwise low-lying county.

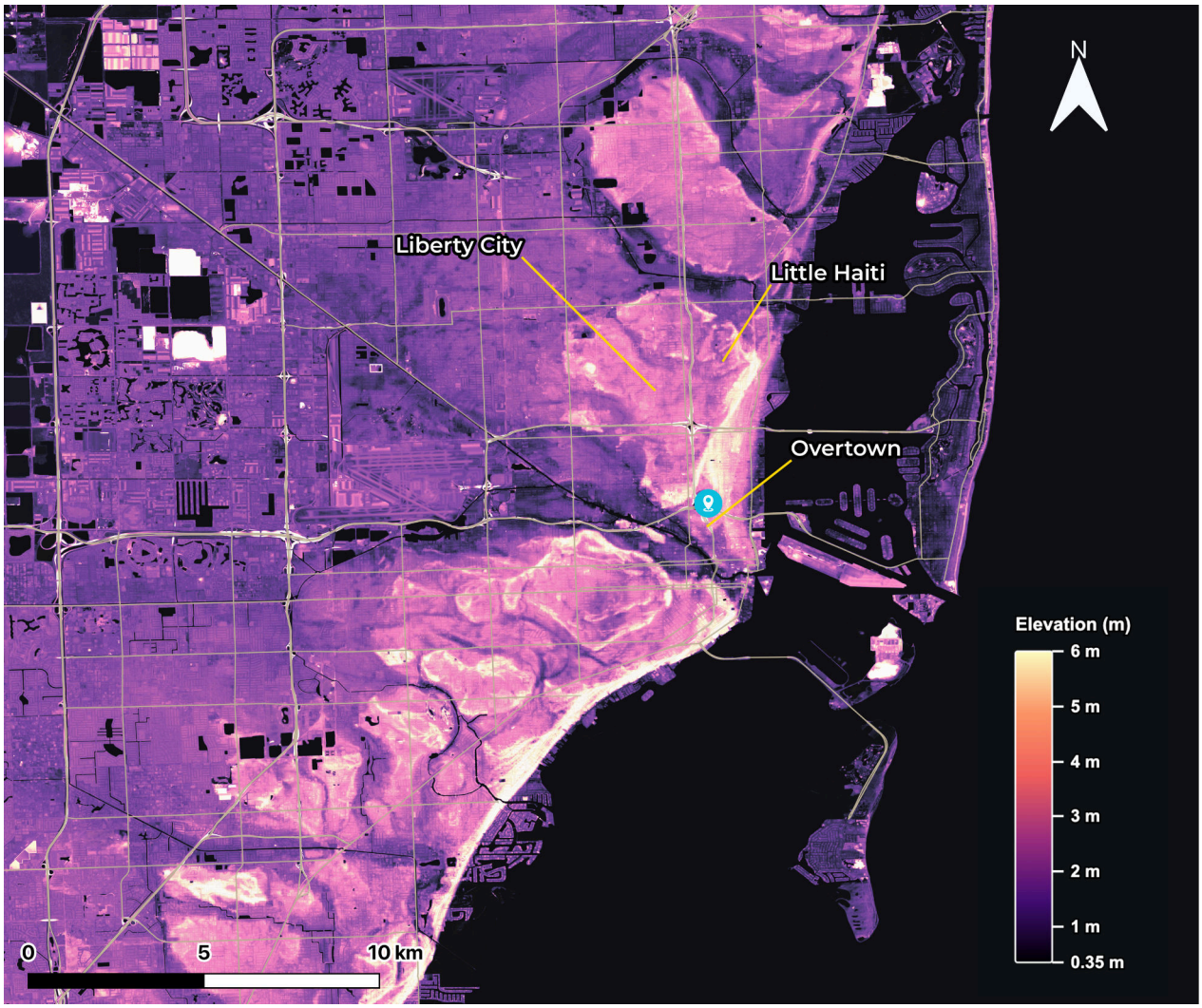


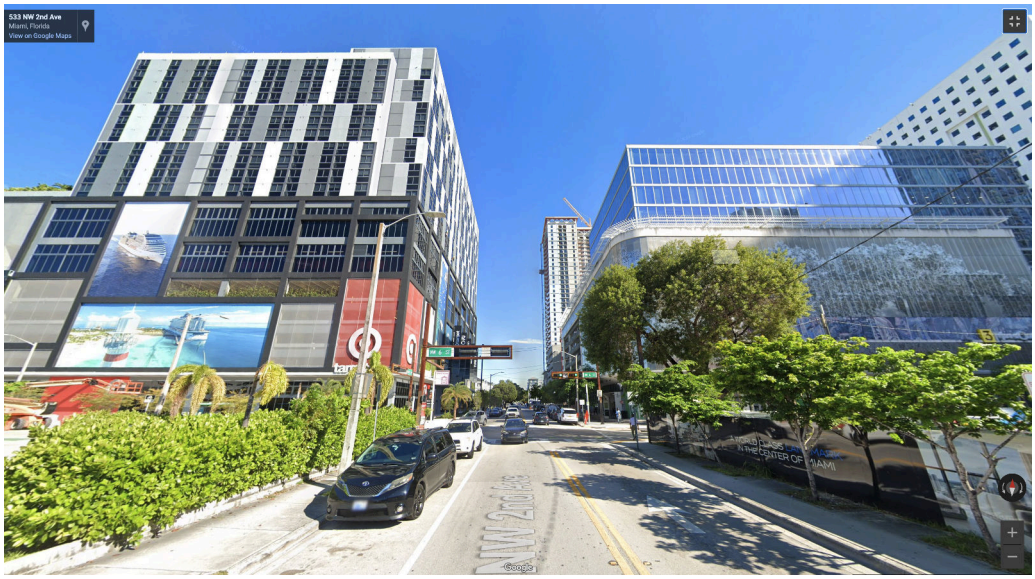
Figure 2. Digital Elevation Model of the Atlantic Coastal Ridge within east–central Miami-Dade County showing the topographical prominence of Overtown, Liberty City and Little Haiti relative to the surrounding low-lying terrain. The cyan marker shows the location of Figure 3.

The remainder of the paper is structured as follows. Section 1.1 provides an overview of the literature on climate gentrification and displacement, which is followed by a description of the significance of this study in Section 1.2. The data and analysis methods used in

this research are described in Section 2. Results are presented in Section 3 and discussed in Section 4. Finally, Section 5 concludes the work and provides directions for future research.



(**A**)



(**B**)

Figure 3. Street-level evidence of gentrification in Overtown at the intersection of NW 2nd Ave and NW 6th St (Cyan marker in Figure 2); (**A**) the area in 2011 showing the neighborhood's historically low-investment commercial fabric. (**B**) The same location in 2024 with new mixed-use construction and visible changes in the built environment consistent with accelerating real estate investment.

1.1. Literature Review

As climate change intensifies the frequency and severity of coastal flooding, physical climate risk is increasingly incorporated into property market valuations. Research in coastal housing markets has demonstrated that properties within flood hazard zones trade at significant discounts relative to comparable properties outside designated risk areas [9] and that flood experience reshapes risk perceptions and subsequent market behavior in affected communities [10]. However, this repricing is incomplete: flood-prone residential properties in the United States are collectively estimated to be overvalued by USD 121–237 billion [11], suggesting that market correction is ongoing and that further displacement

pressures will intensify as insurance costs and risk awareness increase. The counterpart to this devaluation of low-lying areas is the growing premium commanded by higher-elevation, flood-resilient land. In Miami-Dade County, Keenan et al. [3] demonstrated that the rate of property value appreciation between 1971 and 2017 was positively and significantly associated with elevation, which is a pattern they termed the elevation hypothesis. The repricing of elevation as a climate resilience asset thus operates simultaneously on two fronts: depressing values in flood-exposed areas while inflating them in areas perceived as safe. It is within this dual dynamic that the conditions for climate gentrification emerge.

The mechanism of climate gentrification is an extension of classical gentrification theory: as the exchange value of high-elevation land increases, residents who cannot capitalize on that appreciation face displacement through rent increases, property tax burdens and speculative redevelopment pressure. Renters are especially exposed: in Overtown, 90% of occupied housing units were renter-occupied in 2023 (84% in 2010), and 77% across Overtown, Liberty City, and Little Haiti (68% in 2010), compared with 46% across all high-elevation areas (40% in 2010) [12]. Appreciation therefore accrues to owners and landlords rather than residents. Taylor and Aalbers [13] examine the financial architecture underlying this process and demonstrate how property finance actors including insurers, lenders and developers are actively reorganizing urban space along climate risk lines. Comparative evidence from a different coastal context supports the generalizability of the pattern. It was found in Orleans Parish, Louisiana following Hurricane Katrina that gentrification was directly associated with higher ground elevation and inversely associated with flood depth and that census tracts eligible for gentrification were disproportionately Black with 80% Black population shares in 2000 [14]. Climate gentrification is therefore not merely an economic phenomenon but also a racial one. It extends rent-gap theory to climate risk [13] and has been framed as a coupled human and natural system [15], while neighborhood change can occur through succession as well as displacement [16].

Evidence from other coastal regions is growing but remains uneven. Best et al. [17] clustered 51 US east coast counties and identified different pathways, including investment in low-exposure counties and disinvestment in high-exposure counties, with Miami-Dade grouped in the latter at the county scale. Their tract-level case study in North Carolina showed variation that the county classification obscured, indicating that conclusions depend on spatial scale. A review of U.S. studies found that most examine single locations and identified gaps in longitudinal data, which is a measurement of displacement and causal inference [15]. Simulation work further suggests that repeated flooding can shift housing market composition and leave lower-income households in hazard zones [18]. Outside the United States, a study of planned relocation in Tacloban City, Philippines, found that climate gentrification there was shaped by politics, access to capital and livelihoods as well as hazard exposure [19]. These studies show that the mechanisms of climate displacement vary with local housing markets, hazard exposure, and settlement history, which motivates testing the Miami-Dade pattern across spatial scales. A related theory, green or environmental gentrification, explains how sustainability investments, including flood resilience projects, raise neighborhood desirability and displace the communities they are supposed to serve [20,21]. While the mechanisms of climate and green gentrification differ, both studies agree that the spatial unevenness of climate risk and climate adaptation generates new frontiers of displacement in cities where low-income communities of color are disproportionately concentrated in areas that are either newly hazardous or newly desirable. In the context of this study, the focus areas are more desirable due to their higher elevation, as described in Section 1.

The spatial pattern of climate gentrification risk in MDC reflects a direct inheritance of racially exclusionary housing policy. The concentration of Black non-Hispanic residents

along the Miami Rock Ridge traces to a history of legally enforced segregation that confined Black residents to specific corridors of the urban landscape away from the waterfront areas that dominated Miami's development [8]. Mohl [6] documents how racial zoning, redlining, segregated public housing, and urban renewal and highway construction sustained this separation across the 20th century (Figure 4). Under SLR, this same high-elevation land has become a climate asset, and its rising market value is now displacing the communities that historical exclusion confined there. This indirect displacement mechanism is documented by Seeteram et al. [22], who found that under a 1 m SLR scenario in MDC, 56% of residents in low-flood-hazard areas face economic displacement driven not by direct inundation but by market pressures generated as higher-hazard residents relocate. The regional adaptation planning framework has recognized the physical dimensions of this risk. The Southeast Florida Regional Climate Compact and Miami's Resilient305 strategy have established elevation-based risk stratification as a core planning principle, identifying low-lying coastal communities as primary targets for resilience investment while leaving implicit the distributional consequences for adjacent higher-elevation communities [23,24], yet adaptation plans have been slower to address the equity implications of the spatial sorting they implicitly endorse [25].

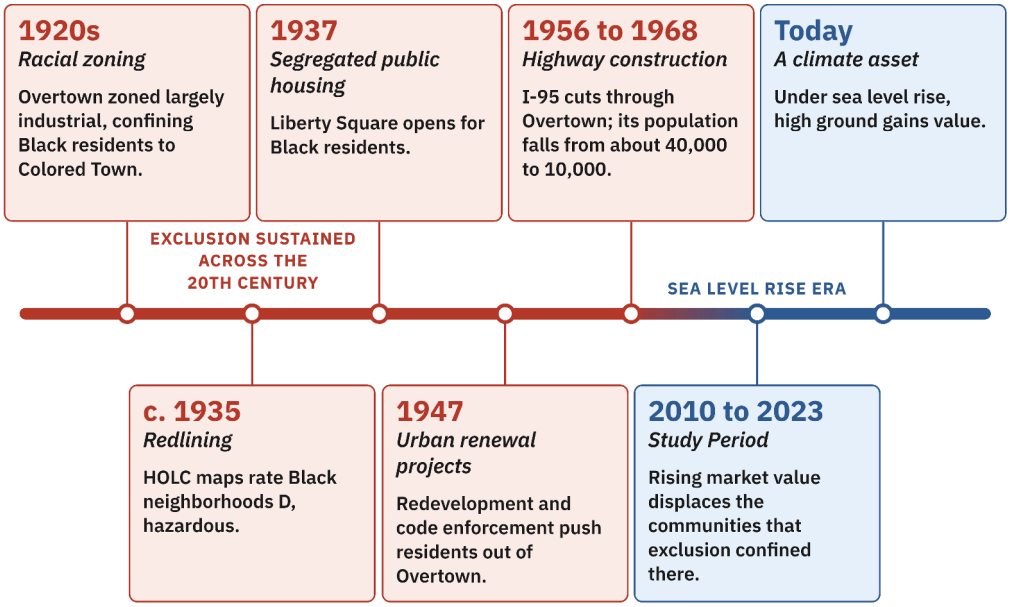


Figure 4. Policy mechanisms that confined Black residents to the Miami Rock Ridge and the shift in the ridge's value under SLR. Mechanisms follow Mohl [6]; dates and the Overtown population change are drawn from [26,27].

1.2. Significance of This Study

Many empirical studies examining the relationship between climate risk and urban change operate at scales too broad to detect neighborhood-level dynamics [3,28]. Global regression models implicitly assume that the relationship between environmental exposure and socioeconomic change is spatially stationary, that is, constant across the study area. In practice, this assumption is frequently violated. Spatial heterogeneity, where the relationship between two variables (such as elevation and property appreciation) varies by geographic location, can render county-wide coefficient estimates misleading or even directionally incorrect at the local scale.

This is especially important in a study of climate gentrification. If displacement pressure is geographically concentrated, that is, clustered around specific elevations or specific neighborhoods, a global model will systematically underestimate the magnitude of the

effect in the areas where it is strongest and may fail to detect it entirely. This study addresses that limitation directly by combining a county-wide regression framework with a spatially explicit sub-analysis focused on the Miami Rock Ridge, which is the geographic feature that links climate resilience, historical segregation and present-day displacement pressure in MDC. By doing so, we provide a neighborhood-scale empirical assessment of climate gentrification that is grounded in the county's specific physical and social geography.

Prior empirical work has evaluated climate gentrification through the relationship between elevation and single-family property price appreciation [3], through threshold-based tract eligibility and a gentrification index analyzed with ordinal logistic regression [14] and through the county-scale clustering of multiple vulnerabilities [17]. None of these studies combine a demographic displacement indicator with global and local regression across nested spatial scales. This study addresses that gap with four goals: (1) to test demographic displacement alongside income and home value change, (2) to determine whether the elevation–displacement relationship is spatially stationary or varies locally, (3) to isolate the ridge communities where historical marginalization and climate resilience coincide, and (4) to test whether the relationship persists among tracts already above the planning elevation threshold.

2. Materials and Methods

This study analyzes 646 census tracts across MDC, Florida from 2010 to 2023, integrating social, demographic and environmental data to examine the spatial dynamics of climate gentrification. The study area is shown in Figure 1. This section describes the data sources and preparation (Sections 2.1 and 2.2), the delineation of the ridge communities (Section 2.3), the four-stage analytical framework (Section 2.4) and the robustness and sensitivity analysis (Section 2.5). Abbreviations are defined in full where first introduced, and a complete list is also provided in the Abbreviations section at the end of the manuscript.

2.1. Data Sources

Demographic and economic indicators were collected from the U.S. Census Bureau's American Community Survey (ACS) 5-year estimates for 2010 and 2023. The 2010 release spans the 2006–2010 period and is anchored to 2010 tract boundaries; the 2023 release spans 2019–2023 and is anchored to 2020 tract boundaries, yielding a 13-year comparison between non-overlapping 5-year periods, as the Census Bureau recommends [29]. The 2010 release ends as the foreclosure crisis in Miami-Dade peaked [30], and the 2023 release covers the post-COVID-19 period, which brought pandemic-driven migration [31] and a surge in Florida home prices [32]. All ACS estimates were retrieved via the Census API for MDC. Median household income serves as the primary economic indicator, as it is consistently available across the full time series and captures neighborhood-level economic shifts through tract aggregation rather than individual-level variation. Median home value, Black non-Hispanic population and total population were also retrieved. Median home value appreciation was computed as the percentage change between the 2010 and 2023 ACS estimates and serves as a complementary market indicator alongside income change, capturing shifts in asset values that may precede or accompany household-level economic transitions. The choice of Black non-Hispanic population as the demographic indicator of displacement reflects the specific socioeconomic composition of MDC. Approximately 70% of the county identifies as Hispanic, but this category cuts across racial lines and includes White, Afro-Latino, Indigenous and multiracial individuals; therefore, a raw Hispanic population count would mask the racial heterogeneity within that group and obscure neighborhood-level demographic shifts. Black non-Hispanic population share provides a

more precise indicator of the communities historically segregated on the Miami Rock Ridge and have remained beyond the abolition of such exclusionary policies.

The percentage of each census tract inundated under a 2.1 m (7 ft) SLR scenario was derived from the NOAA Office for Coastal Management Sea Level Rise inundation dataset [7], which models stillwater inundation extent above a NAVD88 baseline using a hydrostatic "bathtub" approach which flags areas below a projected water level that are hydrologically connected to the coast and applied to a high-resolution coastal DEM [33,34]. The selection of the 2.1 m threshold is grounded in the regional planning framework described in Section 2.3. The inundation polygon was intersected with census tract boundaries, and the proportion of each tract's area falling within the inundated extent was computed. This variable serves as a tract-level proxy for chronic SLR flood exposure. Elevation data were derived from a 2021 5-foot (1.5 m) resolution Digital Elevation Model (DEM) provided by Miami-Dade County [35], from which mean tract elevation was calculated using zonal statistics. Table 1 summarizes all data sources and variable codes.

Table 1. Data sources and variables. Code refers to the specific ACS fields that were used.

Variable	Source	Code	Years
Median Household Income	ACS 5-yr	B19013_001E	2010, 2023
Median Home Value	ACS 5-yr	B25077_001E	2010, 2023
Black Non-Hispanic Population	ACS 5-yr	B03002_004E	2010, 2023
Total Population	ACS 5-yr	B01003_001E	2010, 2023
Mean Tract Elevation	2021 DEM	—	2021
Percentage inundated (2.1 m SLR)	NOAA	—	2024

2.2. Data Preparation and Harmonization

Because census tract boundaries changed between 2010 and 2020, a direct comparison of 2010 and 2023 ACS estimates would introduce geometric measurement error wherever tract boundaries were redrawn. To address this, 2010 tract estimates were harmonized to 2020 geometries using the National Historical Geographic Information System (NHGIS) population-weighted crosswalk [36]. This procedure splits each 2010 tract proportionally into its corresponding 2020 tracts using population weights (`wt_pop`); then, it reaggregates the resulting values to produce estimates on a consistent 2020 geometry. The population-weighted interpolation of this kind is standard for harmonizing census geographies over time [36,37]. The NHGIS weights are derived from block-level populations, which captures the within-tract variation in where people live. Dasymetric approaches that use ancillary data such as land cover or parcels can refine such estimates further [38], but they require additional inputs that are not available consistently for both periods. Counts were reallocated directly, and medians were approximated as weighted averages.

All spatial layers were projected to NAD83(HARN) / Florida East (EPSG:2881; US survey feet) to support accurate distance calculations. Integrated ACS data, SLR inundation ratio and DEM-derived elevation were joined to 2020 tract geometries in a single working dataset; then, they were imported to a file geodatabase for subsequent spatial analysis. Elevation values were referenced to the North American Vertical Datum of 1988 (NAVD88). Mean elevation per tract was computed from the DEM using zonal statistics, and geodesic distance from each tract centroid to downtown Miami (Flagler Street and Miami Avenue) was calculated using the Near tool.

A two-step outlier filter was applied before analysis. First, tracts with a 2010 baseline median household income below USD 10,000 were excluded, as values this low typically reflect industrial, non-residential or institutional populations (e.g., correctional facilities) rather than genuine residential communities. Tracts with an income change exceeding ±300% were also excluded, as swings of this magnitude are more consistent with data

artifacts rather than real socioeconomic change. These filters reduced the working sample from 707 to 646 census tracts.

2.3. Vulnerable Ridge Community Delineation

The three communities mentioned in Section 1 (Overtown, Liberty City and Little Haiti) share two defining characteristics: historical disinvestment rooted in 20th century segregation and physical location that sits at higher elevations than the surrounding landscape (Miami Rock Ridge). Capturing the intersection of physical climate resilience and socioeconomic vulnerability requires screening on both dimensions. Following prior gentrification-eligibility frameworks that apply multiple tract-level criteria [14,16], we use a two-tier delineation: a physical elevation filter followed by baseline income and demographic filters.

First, the communities residing on the physical ridge are approximated using a mean tract elevation threshold at or above 2.1 m NAVD88 (7 feet), identifying a high-elevation baseline of 277 tracts. This elevation threshold is grounded in both the empirical elevation distribution of MDC, where 2.1 m sits immediately above the county-wide mean tract elevation of 2.07 m (6.79 ft) and the official SLR planning framework for southeast Florida. The Southeast Florida Regional Climate Change Compact's Regionally Unified Sea Level Rise Projection, which was reaffirmed as the authoritative regional planning standard in December 2024 [23,39], defines a 50-year (2070) planning range of 0.53–1.02 m for general infrastructure and 1.37 m for critical assets. These correspond to the NOAA Intermediate–High and High scenarios [40]. These scenarios project approximately 1.5 and 2 m, respectively, by 2100 relative to a 2000 baseline [40]. At the closest permanent NOAA tidal gauge, Virginia Key (Station 8723214; https://tidesandcurrents.noaa.gov/datums.html?id=8723214&datum=NAVD88&units=0&epoch=0 (accessed on 15 August 2026) , the tidal surface used as the reference datum for SLR inundation modeling (Mean Higher High Water, MHHW) [33], sits approximately 7 cm (0.23 ft) above 0 m elevation in NAVD88. Adding 2.0 m of century-scale High SLR to MHHW yields a stillwater inundation surface of 2.07 m (6.79 ft) NAVD88. Because NOAA's SLR inundation data are distributed in discrete 1-foot (0.30 m) increments [7], 2.1 m (7 ft) represents the lowest discrete increment above the century-scale High stillwater inundation envelope and is therefore used in this study as the SLR threshold.

We note that future SLR projections carry substantial scenario and process uncertainty arising from post-2050 ice-sheet dynamics, ocean circulation variability in the Florida Current and nonlinear compound flooding from storm surge and extreme precipitation [40]. Furthermore, hydrostatic ("bathtub") inundation models only capture surface connectivity above MHHW [33] and omit groundwater table shoaling through Miami-Dade's porous limestone aquifer, storm-drain backflow and DEM vertical uncertainty [2,33,34]. Consequently, 2.1 m is treated here as a benchmark planning heuristic rather than a deterministic flood boundary. To ensure our findings do not depend on this single cutoff, we evaluate sensitivity across elevation thresholds both in the gradient analysis (Section 2.4.2) and the robustness checks (Section 2.5).

Second, to isolate historical socioeconomic vulnerability, two baseline demographic filters are applied to the high-elevation tracts:

1. 2010 median household income below USD 40,000, targeting tracts economically disadvantaged at the start of the study period. This threshold approximates the 40th percentile of 2010 tract median household income in MDC (USD 37,708), which is the income criterion commonly used to identify gentrification-eligible tracts [16].
2. 2010 Black non-Hispanic population share above 30%, capturing the historically Black communities that segregation-era policy placed on this high ground [6]. Studies

commonly classify neighborhoods as majority Black when more than half of residents are Black [41]. A lower floor of 30% was used here to capture tracts with a substantial Black population—not only majority-Black tracts. The median baseline share among VRC tracts is 71.8%.

Applying these socioeconomic filters subdivides the 277 high-elevation tracts into Vulnerable Ridge Communities (VRCs, $n = 42$) and non-VRC ridge tracts ($n = 235$). Figure 1 shows the location of the 42 VRC tracts within MDC. Note that tracts on the ridge with higher incomes or different demographic composition are explicitly excluded, as the selection criteria target the intersection of physical climate resilience and historical socioeconomic vulnerability.

2.4. Analytical Framework

The analysis proceeds in four stages, each corresponding to the section of the Results presented in Section 3. Figure 5 illustrates the entire workflow.

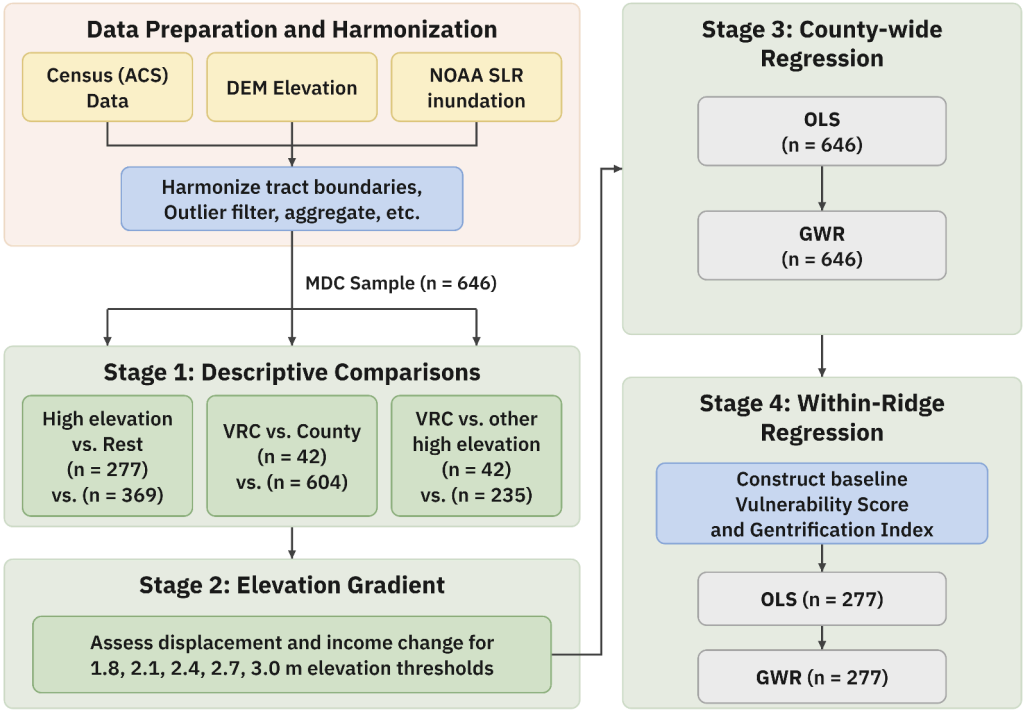


Figure 5. Data analysis workflow.

2.4.1. Stage 1: Descriptive Comparisons

The first stage (Section 3.1) involves descriptive comparisons utilizing non-parametric tests to establish baseline differentials across three gentrification indicators: percent change in median household income, percent change in median home value and change in Black non-Hispanic population share. Three primary comparisons are evaluated:

- All high-elevation tracts ($n = 277$) versus low-elevation tracts ($n = 369$) within MDC to isolate the effect of elevation alone.
- VRC ($n = 42$) versus the rest of MDC baseline ($n = 604$) to establish the overall magnitude of the differential in gentrification and displacement between the target communities and the county average.
- VRC ($n = 42$) versus non-VRC ridge tracts ($n = 235$) to isolate the effect of historical socioeconomic marginalization.

Continuous variables are tested using the Mann–Whitney U-test, while binary indicators (e.g., presence of new construction) are evaluated using Chi-square and Fisher's exact tests.

2.4.2. Stage 2: Elevation Gradient Analysis

The second stage (Section 3.2) examines how displacement and income vary with elevation. At each of five elevation thresholds (1.8, 2.1, 2.4, 2.7, 3.0 m) that correspond to 6–10 feet, tracts meeting that threshold alongside the income and demographic criteria are identified. The median Black non-Hispanic population change and change in median household income are computed for the resulting sub-samples. Comparing these values across thresholds produces an elevation gradient profile that serves as a sensitivity analysis further supporting the empirical meaningfulness of the 2.1 m threshold relative to adjacent cutoffs and reveals whether displacement and investment pressures are concentrated at the same elevation or diverge spatially along the ridge.

2.4.3. Stage 3: County-Wide Regression Analysis

The third stage (Section 3.3) applies county-wide regression to assess the association between physical climate exposure and two gentrification outcomes across all 646 tracts. Two parallel Ordinary Least Squares (OLS) models were estimated with identical predictors but different dependent variables: the percentage change in median household income and the percentage-point change in Black non-Hispanic population share (2010–2023). Predictors in both models include the percentage of each tract inundated under the NOAA 7-foot (2.1 m) SLR scenario (`pct_inund`), mean tract elevation (m), distance to the Miami CBD (km) and 2010 baseline median household income (USD 10K units). The Black non-Hispanic displacement model additionally includes the 2010 baseline Black non-Hispanic population share as a compositional control, since percentage-point changes are mechanically bounded by the baseline share and its omission would confound the elevation coefficient. Because of this compositional bound, we also estimate a specification including a centered elevation $\times$ baseline Black non-Hispanic share interaction term to test whether the elevation effect depends on initial demographic concentration. The predictors follow prior climate gentrification research, in which elevation and flood exposure are the physical exposure measures and baseline income and Black population share are core covariates [14,42,43], and distance to the Miami CBD controls for centrality.

The Variance Inflation Factor (VIF) was computed for all predictors to screen for multicollinearity. Residual normality (Jarque–Bera, Shapiro–Wilk), heteroscedasticity (Breusch–Pagan, White) and spatial dependence (Moran's I and Lagrange multiplier lag and error tests) were assessed for all OLS models (Table A1). Heteroscedasticity-robust (HC3) standard errors [44] were computed alongside conventional standard errors, and OLS residuals were examined for spatial autocorrelation using global Moran's I with a queen contiguity weights matrix.

Significant residual clustering indicated spatial dependence. Spatial lag and spatial error models were estimated as global alternatives, but Geographically Weighted Regression (GWR) [45] was retained because it also allows coefficients to vary across space (Table A2). GWR was estimated using the `mgwr` Python package (version 2.2.1; Python 3.11) with an adaptive bisquare kernel and bandwidth selected by minimizing AICc [46]. GWR produces tract-level local coefficients and local R^2 values, enabling the identification of where the elevation–displacement relationship is spatially concentrated across the county.

2.4.4. Stage 4: Within-Ridge Modeling

The fourth stage (Section 3.4) applies spatial modeling to the high-elevation tracts ($n = 277$). To support continuous spatial regression analysis, a baseline vulnerability score

(VS) is defined for each tract. This metric standardizes and combines the 2010 baseline economic and demographic conditions:

$$VS_i = z(-\text{Income}_{2010}) + z(\text{BlackNH}_{2010}) \quad (1)$$

where VS_i is the continuous baseline vulnerability score of tract i, $z(\cdot)$ represents standardized z-scores, Income_{2010} is the 2010 median household income and BlackNH_{2010} is the Black non-Hispanic population share. This equation constructs a composite predictor by placing baseline economic and demographic conditions on a common scale. Higher VS values indicate greater baseline socioeconomic vulnerability.

A Gentrification Intensity Index (GI) is also constructed to jointly capture gentrification-driven displacement as follows:

$$GI_i = z(\Delta\text{Income}_i) - z(\Delta\text{BlackNH}_i) \quad (2)$$

where $z(\cdot)$ denotes standardization to zero mean and unit variance. ΔIncome_i is the percent change in median household income and $\Delta\text{BlackNH}_i$ is the percentage-point change in Black non-Hispanic population share for tract i. Standardizing both terms places them on a common scale so that each contributes equally to the index, which is consistent with composite social status indices that average their components and define gentrification as the change in that index over time [47]. Higher values of GI indicate tracts experiencing simultaneous income growth and decline in Black non-Hispanic share, which is consistent with gentrification as reinvestment accompanied by demographic succession [16].

Subsequently, an OLS model was estimated to assess which factors influence GI. The model uses four predictors: percentage of tract area inundated under 2.1 m SLR, mean elevation, distance to Miami CBD and the continuous baseline vulnerability score (VS). Global Moran's I was computed on the OLS residuals to assess spatial autocorrelation utilizing a k-nearest-neighbor weights matrix ($k = 5$, row-standardized) rather than queen contiguity to account for the spatial fragmentation of the ridge tracts. Finally, a GWR model was also estimated on the high-elevation tracts utilizing the same set of predictors, an adaptive bisquare kernel and bandwidth selected by minimizing AICc.

2.5. Robustness and Sensitivity Analysis

Five sets of sensitivity analyses were conducted across the analytical stages to verify that findings are not artifacts of threshold selection, compositional bounds, bandwidth choice, or index construction (reported in Appendix A):

1. VRC Delineation Thresholds (Stage 1; Table A7): To assess classification bias in the VRC definition, Mann–Whitney U comparisons between VRC and non-VRC ridge tracts were repeated across all nine combinations of baseline median household income (USD 35,000, USD 40,000 and USD 45,000) and Black non-Hispanic share (25%, 30% and 35%) cutoffs. Home value appreciation was significantly higher in VRC tracts under all nine combinations ($p < 0.001$), and median Black non-Hispanic share declined by 14.7 to 17.4 percentage points. Income growth was significantly higher in VRC tracts at the 35,000 and 40,000 cutoffs ($p = 0.004$ to 0.035) but attenuated at USD 45,000 ($p = 0.257$ to 0.263), as raising the cutoff well above the 40th county percentile (USD 37,708) dilutes the sample with middle-income tracts.
2. Baseline Compositional Subsamples (Stage 3; Table A3, Panel B): To confirm that county-wide displacement results are not driven by larger baseline populations having more percentage points to lose, the county-wide Black non-Hispanic OLS model was re-estimated within subsamples restricted to tracts with $\geq$ 15% ($n = 174$), $\geq$ 30%

($n = 128$, the VRC threshold) and $\geq 50\%$ ($n = 100$) baseline Black non-Hispanic share, using both percentage-point change and percentage change relative to baseline.

3. County-Wide GWR Bandwidths (Stage 3; Table A4): To ensure that the spatial clustering of local elevation coefficients in northern MDC is not an artifact of the AICc-optimal bandwidth ($bw = 152$), the county-wide GWR was re-estimated across fixed adaptive bandwidths from 60 to 646 nearest neighbors as well as without the baseline demographic control. The concentration of significant negative elevation coefficients held across all bandwidths.
4. Within-Ridge Elevation Cutoffs (Stage 4; Table A5): Because the 2.1 m (7 ft) boundary is a regional planning heuristic rather than a deterministic flood threshold, the within-ridge OLS regression was re-estimated across high-elevation tract definitions ranging from 1.50 m ($n = 539$) to 3.00 m ($n = 68$). Baseline vulnerability (VS) remained highly significant across all cutoffs ($p < 0.001$), and mean elevation remained positive and significant from 1.50 m through 2.70 m ($p = 0.001$ to 0.028), losing significance only at 3.00 m ($p = 0.223$) where the sample size is small ($n = 68$).
5. Gentrification Intensity Index Construction (Stage 4; Table A6): To test sensitivity to equal z-score weighting in the GI composite, the within-ridge OLS model was re-estimated using five alternative dependent variables: income change only, Black non-Hispanic change only, asymmetric component weights (0.5 and 2.0 on Black non-Hispanic change) and a non-parametric rank-based index. Baseline vulnerability (VS) remained a significant predictor across all six variants ($p \leq 0.005$), whereas mean elevation was significant under equal weights ($p = 0.001$) and double demographic weighting ($p = 0.006$) but not in single-component or rank-based specifications.

3. Results

The results follow the four stages of the framework described in Section 2.4. We first compare the Vulnerable Ridge Communities (VRCs) with the rest of the county; then, we test whether elevation is associated with income, home value and Black non-Hispanic population change county-wide. GWR then maps where that relationship varies, and a final within-ridge model examines gentrification intensity among high-elevation tracts.

3.1. Descriptive Analysis

The 42 VRC tracts (Figure 1, right panel) represent approximately 6.5% of the county's 646 census tracts. The 2010 baseline characteristics of these tracts reflect the legacy of segregation in MDC. Their median household income was approximately $28,000, roughly 65% of the county median at the time, and their median Black non-Hispanic population share was 71%. The mean elevation of ridge community tracts was 2.88 m (9.45 feet), while the mean elevation of the county was 2.07 m (6.79 feet).

VRCs experienced socioeconomic and physical changes substantially exceeding the non-ridge county baseline (Table 2). Median household income grew by 81.7% in VRC tracts compared to 58.3% across the rest of the county. A non-parametric Mann–Whitney U-test confirmed that this difference is statistically significant ($p = 0.036$). Median home values appreciated by 85.9% versus 51.7% ($p < 0.001$). To isolate the response to physical climate resilience, all high-elevation tracts (≥ 2.1 m, $n = 277$) were compared against the remaining low-elevation tracts ($n = 369$). High-elevation tracts saw significantly higher home value appreciation (60.7% vs. 47.1%, $p < 0.001$). However, there was no statistically significant difference in median income growth (60.3% vs. 57.3%, $p = 0.667$) or Black non-Hispanic population change (-0.3 pp vs. 0.0 pp, $p = 0.177$).

To assess the role of historical socioeconomic marginalization, the 42 VRC tracts were compared to the remaining 235 high elevation tracts. Despite both these groups being

located on the ridge (mean elevation 2.88 m (9.45 ft) vs. 2.70 m (8.86 ft)), gentrification outcomes differed. Income growth was significantly higher in VRCs (81.7% vs. 59.5%, $p = 0.031$), as was home value appreciation (85.9% vs. 54.4%, $p < 0.001$). There was a 15.0 percentage point decline in Black non-Hispanic population in VRCs; however, this is attributable to the selection criteria.

Table 2. Descriptive comparisons of median gentrification indicators (2010–2023).

Indicator	VRC ($n = 42$)	Rest of County ($n = 604$)	Other Ridge ($n = 235$)	Low-Elev ($n = 369$)
Median Income Growth (%)	81.7	58.3	59.5	57.3
Median Home Value Apprec. (%)	85.9	51.7	54.4	47.1
Median Black NH Change (pp)	−15.0	0.0	0.0	0.0

3.2. Elevation Gradient Analysis

While the ridge community sub-sample was defined at the 2.1 m (7 feet) threshold, we also examine how gentrification indicators vary across the elevation spectrum to characterize the spatial structure of climate gentrification pressure. Tracts were grouped into five elevation bands from 1.8 to 3.0 m (6 to 10 feet). Two indicators were computed for each band and compared to the county-wide average. The indicators were median household income growth and demographic displacement (Black non-Hispanic population change) between 2010 and 2023.

Figure 6 shows that both indicators intensify at higher elevations. Demographic displacement (Figure 6A) shows a decline in the Black non-Hispanic population, dropping from −15.0 percentage points at the 2.1 m band to roughly −20 points at the 2.7 and 3.0 m bands (−20.3 at 2.7 m). At the same time, income growth (Figure 6B) climbs from 82% to over 105% at the 3.0 m peak.

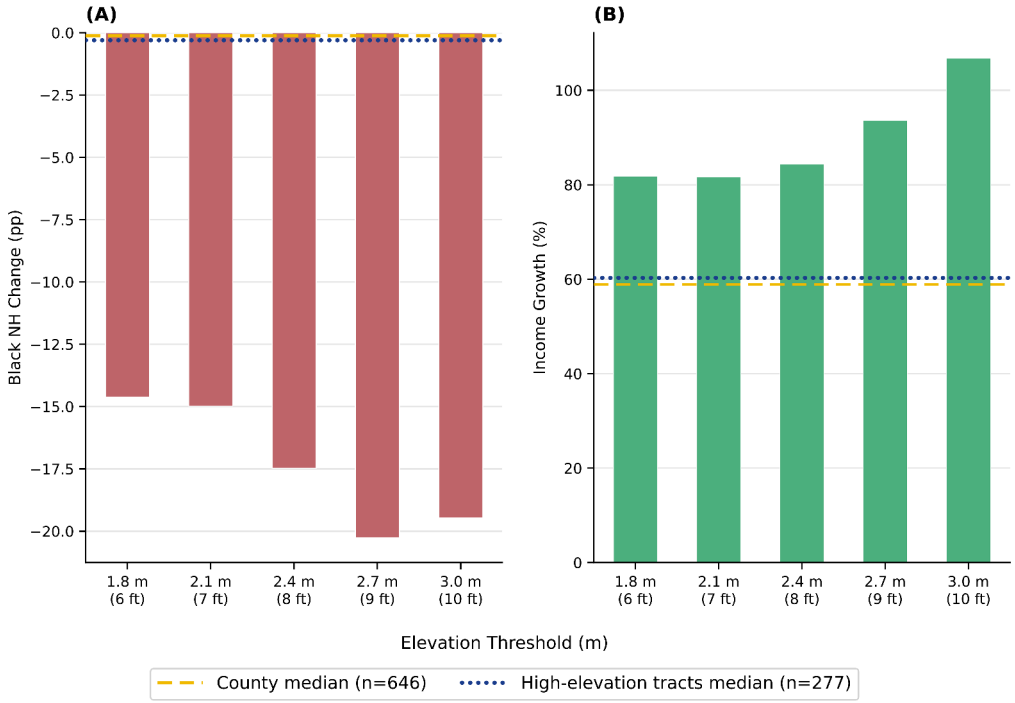


Figure 6. Elevation gradient analysis across five threshold bands (1.8–3.0 m). (**A**) Median change in Black non-Hispanic population share (pp, 2010–2023); (**B**) median household income growth (%, 2010–2023). Dashed yellow lines indicate the county-wide average, while dotted dark blue lines indicate the median values of all high elevations tracts (≥2.1 m/7 ft) for both indicators.

3.3. County-Wide OLS Regression

To assess the relationship between physical climate exposure and neighborhood-level gentrification outcomes at the county scale ($n = 646$), two parallel OLS models were estimated using identical predictors but different dependent variables: (1) the income-change model that assesses the association between the predictors and the percentage change in median household income (2010–2023) and (2) the demographic displacement model that relates the percentage-point change in Black non-Hispanic population share (2010–2023) to the same set of variables. Predictors included the percentage of each tract inundated under the NOAA 7-foot (2.1 m) SLR scenario, mean tract elevation (m), distance to the Miami CBD (km), and 2010 baseline median household income ($10K units). The Black non-Hispanic model additionally controls for the 2010 baseline Black non-Hispanic population share to account for the compositional ceiling on percentage-point change in tracts with lower initial shares. Results are reported in Table 3.

In the income-change model, physical climate exposure variables lacked explanatory power at this scale: neither flood inundation nor mean elevation approached statistical significance. The model was dominated by distance to downtown and baseline income, which is consistent with a pattern of mean-reversion in which lower-income tracts grew faster regardless of climate exposure ($R^2 = 0.175$). In the Black non-Hispanic displacement model, mean elevation was negatively associated with population change ($\beta = -1.516$, $p = 0.013$), but the estimate was not significant under heteroscedasticity-robust standard errors ($p_{HC3} = 0.120$; Table 3). Flood inundation, distance and baseline income were not significant. The weak county-wide estimate reflects demographic geography: the median tract was only 3.3% Black non-Hispanic in 2010. When baseline composition is taken into account, the association emerges. A significant elevation $\times$ baseline share interaction ($p_{HC3} < 0.001$) shows no elevation association in tracts with few Black non-Hispanic residents but a strong one where they were concentrated: each additional meter corresponded to a 1.2 percentage point decline at a 16.1% baseline share ($p = 0.003$) and a 4.9 point decline at 66.9% ($p < 0.001$). Within the 128 tracts that were at least 30% Black non-Hispanic in 2010, each meter corresponded to a 6.6 percentage point greater decline ($p_{HC3} = 0.001$), or 8.8% of the baseline share ($p_{HC3} = 0.020$), so the pattern is not explained by larger populations having more to lose (Table A3).

Table 3. County-wide OLS regression results ($n = 646$). Conventional and heteroscedasticity-robust (HC3) p-values are reported. Climate exposure variables are not associated with income change. Elevation is negatively associated with Black non-Hispanic change under conventional standard errors but not under robust standard errors; Section 3.3 reports the association conditional on baseline composition.

	DV: Income Change (%)				**DV: Black NH Change (pp)**			
Variable	**Estimate**	**SE**	p	p_{HC3}	**Estimate**	**SE**	p	p_{HC3}
Intercept	126.906	15.648	**<0.001**	**<0.001**	3.932	2.143	0.067	0.219
Pct. Inundated (2.1 m/7 ft NOAA)	0.007	0.089	0.940	0.944	−0.013	0.012	0.294	0.444
Mean Elevation (m)	1.020	4.454	0.819	0.845	−1.516	0.610	**0.013**	0.120
Distance to Downtown (km)	−0.196	0.068	**0.004**	**0.011**	−0.004	0.009	0.670	0.728
Baseline HH Income ($10K)	−9.432	0.892	**<0.001**	**<0.001**	+0.101	0.127	0.424	0.219
Baseline Black NH Share (%)			—		−0.201	0.012	**<0.001**	**<0.001**
R^2	0.175				0.351			

Note: Bold values indicate statistical significance at $p < 0.05$.

A global Moran's I test on the Black non-Hispanic displacement model residuals confirmed significant spatial autocorrelation (Moran's $I = 0.168$, $p < 0.001$), indicating that

the county-wide OLS estimates are spatially biased and that locally varying relationships remain. Diagnostic tests (Table A1) showed non-normal, heteroscedastic residuals, so HC3 standard errors are reported. Due to significant residual spatial autocorrelation, a Geographically Weighted Regression (GWR) was subsequently estimated using an adaptive bisquare kernel with bandwidth selected by minimizing AICc (152 neighbors). This allows coefficients to vary across space. The GWR improved local model fit relative to the global OLS with mean local R^2 increasing to 0.565 (range: 0.186 to 0.822). GWR lowered AICc relative to OLS (4301.9 vs. 4457.4) and removed residual spatial autocorrelation (Moran's $I = -0.013$). Spatial lag and error models were also tested (Table A2), but since they assume a global relationship across MDC, the GWR model was kept.

The spatial distribution of the local elevation coefficient (Figure 7) reveals that the county-wide elevation–displacement relationship is not geographically uniform. Significant negative local elevation coefficients ($p < 0.05$) were concentrated in northern Miami-Dade communities, including tracts in the North Miami Beach and Miami Gardens areas, which have historically high baseline Black non-Hispanic population shares. In these communities, higher elevation predicts greater Black non-Hispanic population decline. This pattern held across all bandwidths tested (60 to 300 neighbors; Table A4) and with and without the baseline demographic control. Along the Miami Rock Ridge itself, local elevation coefficients were weaker and less consistently significant, suggesting that the county-scale elevation–displacement signal is driven by compositional dynamics in high-baseline-Black-NH communities in the north rather than by the climate gentrification mechanism concentrated on the ridge. This geographic heterogeneity in the elevation–displacement relationship provides the empirical motivation for the focused within-ridge analysis in Section 3.4, which isolates the climate-specific drivers of gentrification intensity within the high-elevation corridor.

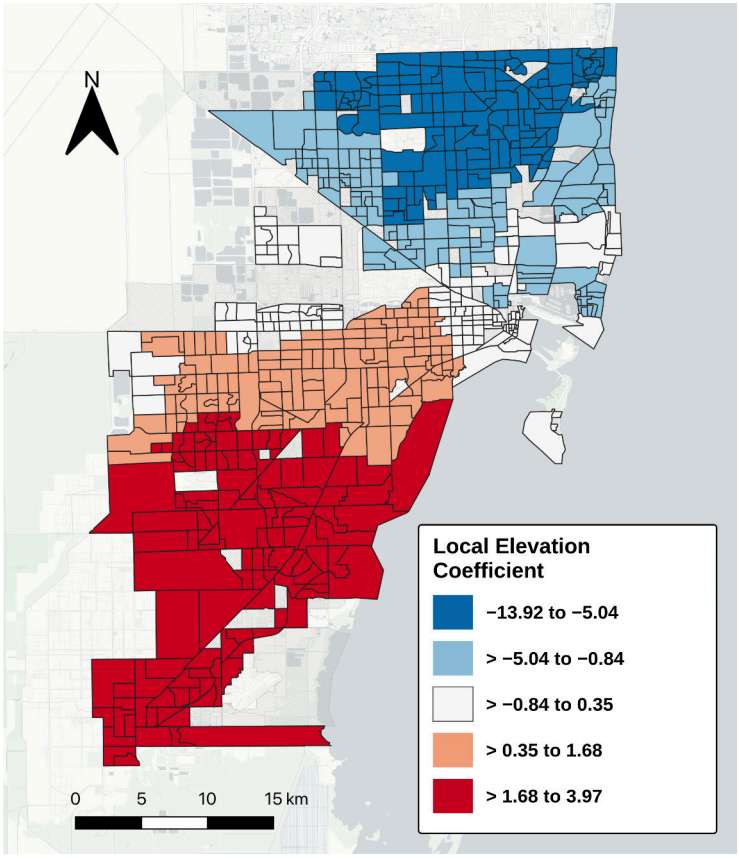


Figure 7. Local elevation coefficient from the GWR predicting Black non-Hispanic population change. Blue tones indicate tracts where higher elevation predicts greater decline.

3.4. Within-Ridge Regression

To evaluate the spatial drivers of compounding displacement pressure across the high-elevation corridor ($n = 277$), an OLS regression was estimated using the composite Gentrification Index (GI) described in Section 2.4.4 as the dependent variable. The global OLS model yielded an adjusted R^2 of 0.318 (Table 4). The baseline vulnerability score was the strongest predictor of gentrification-driven displacement ($\beta = 0.470$, $p < 0.001$). Mean tract elevation also emerged as a significant positive predictor ($\beta = 0.423$, $p = 0.007$) within the ridge, whereas the percentage of tract area inundated under the 2.1 m (7 ft) SLR scenario and distance to downtown were not statistically significant. The vulnerability score was robust to alternative index constructions, but the elevation association was not (Table A6).

Figure 8 maps the clustered spatial distribution of GI across the 277 ridge tracts (Moran's $I = 0.226$, $p = 0.001$), revealing that the highest gentrification intensity is concentrated in the northern ridge communities of Liberty City, Little Haiti and Overtown with more moderate values toward the southern corridor.

Significant spatial autocorrelation remained in the OLS residuals (Moran's $I = 0.205$, $p = 0.001$), indicating unmodeled spatial structure. While global spatial lag and spatial error models partially accounted for this dependence, GWR provided a substantially stronger fit ($\Delta AIC_{OLS} = -29.6$ for spatial error vs. $\Delta AICc_{OLS} = -51.6$ for GWR) and eliminated residual autocorrelation ($I = 0.036$, Table A2), indicating that the residual clustering reflects spatially varying relationships rather than a uniform global spillover. The mean local R^2 increased to 0.487 with local explanatory power ranging from 0.241 to 0.669. Compared to the OLS, AICc also dropped 51.6 units (from 896.4 to 844.8). The local coefficient for baseline vulnerability ranged from 0.369 to 1.013 (mean: 0.744) and remained statistically significant ($p < 0.05$) across 100% of the 277 ridge tracts. These results indicate that while baseline demographic vulnerability universally drives displacement across the ridge corridor, the magnitude of this pressure is spatially heterogeneous.

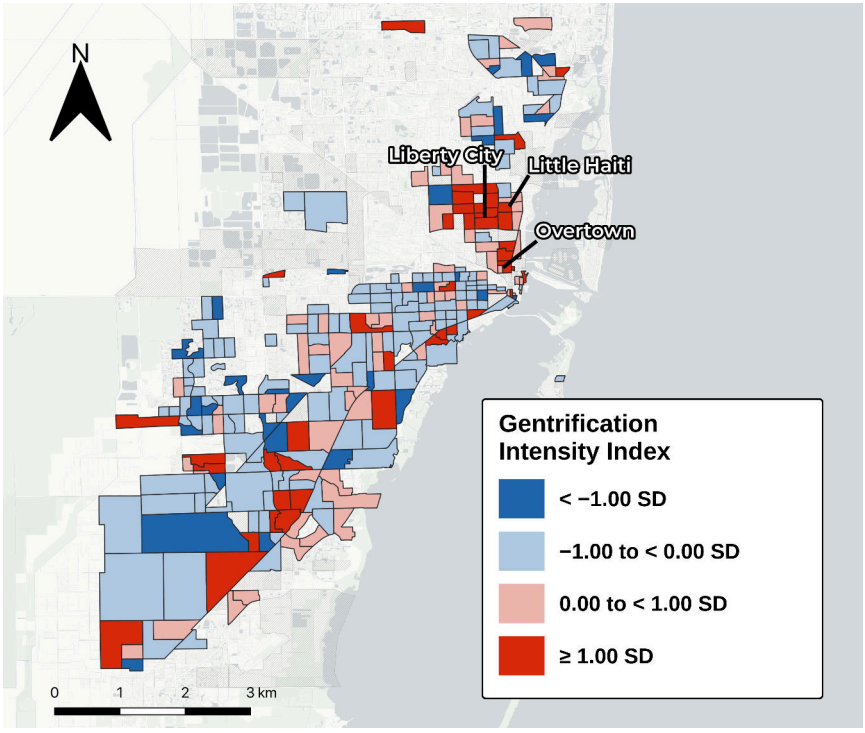


Figure 8. Spatial distribution of the Gentrification Intensity Index (GI) across the 277 high-elevation ridge tracts (2010–2023). Higher values indicate simultaneous income growth and Black non-Hispanic population decline. Liberty City, Little Haiti and Overtown exhibit the most acute gentrification pressure within the ridge corridor.

Table 4. Within-ridge OLS regression results. Dependent variable: Gentrification Intensity Index (GI); $n = 277$; Adjusted $R^2 = 0.318$. Standard errors and p-values are conventional OLS estimates; HC3-based p-values are reported in Tables A5 and A6.

Variable	Coefficient	Std. Error	p-Value
Intercept	−1.100	0.520	**0.035**
Pct. Inundated (7 ft NOAA)	0.005	0.004	0.202
Mean Elevation (m)	0.423	0.155	**0.007**
Distance to Downtown (km)	−0.001	0.003	0.788
Baseline Vulnerability Score (VS)	0.470	0.046	**<0.001**

Note: Bold values indicate statistical significance at $p < 0.05$.

4. Discussion

The most consequential finding of this study is the divergence between the two county-wide models: mean tract elevation showed no association with income growth but a negative association with Black non-Hispanic population change that was conditional on baseline composition. Among tracts at least 30% Black non-Hispanic in 2010, each additional meter of elevation corresponded to a 6.6 percentage point greater decline ($p_{\mathrm{HC3}} = 0.001$), whereas the county-wide estimate was imprecise ($\beta = -1.516$, $p_{\mathrm{HC3}} = 0.120$) because most tracts had small baseline Black non-Hispanic populations. This asymmetry reflects both aggregation bias across heterogeneous high-elevation tracts and the temporal mechanics of neighborhood change. Because the Miami Rock Ridge encompasses both historically marginalized Black neighborhoods and affluent coastal-ridge enclaves (e.g., Coral Gables and Pinecrest), strong county-wide income mean-reversion ($\beta = -9.432, p_{\mathrm{HC3}} < 0.001$) masks localized income gains unless models condition on baseline socioeconomic vulnerability. Moreover, demographic turnover often precedes tract-wide income shifts, as incumbent households in climate-resilient, historically Black neighborhoods are displaced before residential replacement is large enough to register in unconditioned county-wide income regressions. This has a direct implication for measurement: studies that operationalize climate gentrification solely through aggregate county-wide income models (i.e., without conditioning on baseline vulnerability or examining demographic turnover) will systematically underdetect the phenomenon. A similar pattern was documented in New Orleans following Hurricane Katrina, where gentrification was directly associated with higher ground elevation and inversely associated with flood depth specifically within gentrification-eligible, disproportionately Black tracts [14]. In Miami-Dade County, the indirect displacement mechanism identified earlier by Seeteram et al. [22] provides the structural context: as demand concentrates in low-hazard, high-elevation neighborhoods, displacement pressure registers in vulnerable communities even when masked in aggregate county-wide income statistics.

The county-wide GWR reveals both the presence of spatial non-stationarity and the limits of unstratified county-scale local modeling. Significant negative local elevation coefficients (indicating that higher elevation is associated with greater Black non-Hispanic population decline) were concentrated in northern Miami-Dade communities including North Miami Beach and Miami Gardens rather than along the Miami Rock Ridge corridor. Because these northern suburban communities form the largest contiguous concentration of high baseline Black non-Hispanic tracts in the county, modest topographic variation across tracts undergoing broader suburban demographic turnover produces a strong local elevation slope. As opposed to this, along the narrow Miami Rock Ridge, a county-wide bandwidth of 152 neighbors pools historically Black VRC tracts together with adjacent affluent ridge enclaves and low-elevation tracts. This attenuation persists across county-wide bandwidths ranging from 60 to 646 nearest neighbors (Table A4). Thus, while county-wide

GWR successfully diagnoses spatial non-stationarity, interpreting its single-outcome demographic map in isolation risks misattributing climate gentrification to northern suburban compositional shifts while obscuring the finer-scale compounding dynamics (simultaneous income growth and racial turnover) occurring within the ridge corridor. Methodologically, this demonstrates that GWR functions not merely as an endpoint robustness check but also as a diagnostic instrument that reveals when spatial structure mixing requires targeted sub-regional stratification.

Within the high-elevation corridor, baseline demographic vulnerability emerges as the strongest and most consistent correlate of gentrification intensity, remaining statistically significant across all 277 ridge tracts in the GWR with a local coefficient ranging from 0.369 to 1.013. This finding is consistent with elevation alone being insufficient for displacement, which appears concentrated where physical climate resilience and socioeconomic vulnerability intersect. The communities most exposed to climate gentrification are not simply those at high elevation but those that are simultaneously at high elevation and historically poor and Black. This intersection reflects the longstanding history of segregationist housing policy that confined Black residents to the Miami Rock Ridge [6,8], which is land that SLR may now have revalued as a climate refuge. Property finance actors, including insurers, lenders and developers, have been shown to reorganize urban space along climate risk lines, generating what Taylor and Aalbers [13] term "risk rents": value captured from anticipated climate-resilience premiums. Because historical segregation and elevation overlap spatially, the results cannot separate climate-related revaluation from other co-occurring neighborhood processes not captured by distance to downtown, such as targeted urban renewal, transit-oriented rezoning and post-2010 housing market recovery.

Within the ridge, elevation also retains a positive association with gentrification intensity net of baseline vulnerability ($\beta = 0.423$, $p = 0.007$, $p_{HC3} = 0.001$), even though all tracts already exceed the 2.1 m (7-foot) planning threshold. Sensitivity analyses indicate that within-ridge elevation predicts the compounding co-occurrence of income growth and Black non-Hispanic population decline rather than either component in isolation (Table A6). This within-ridge gradient suggests that compounding gentrification pressure responds to continuous variation in elevation rather than to the binary above/below distinction that regional planning frameworks operationalize. While elevation thresholds are administrative delineations of flood safety, the social processes they set in motion (i.e., increased desirability, in-migration and incumbent displacement) are not bounded by it. This finding is consistent with the aggregate elevation–appreciation relationship documented by Keenan et al. [3] in MDC, and it extends the findings from property market outcomes to demographic ones.

Our study employed a four-stage analysis framework, in which each scale resolved a question that the adjacent scale cannot answer, and the full picture of climate gentrification only emerges from their combination. The descriptive analysis established that VRC communities experienced gentrification outcomes substantially exceeding county and non-VRC ridge baselines (Sections 3.1 and 3.2), identifying where the phenomenon is concentrated and who it affects. The county-wide OLS provides the scale at which theory predicts climate exposure should operate (Section 3.3) but also reveals its limits: an analyst relying on the income model alone would conclude that climate exposure is irrelevant at the county scale and that gentrification dynamics are driven entirely by urban accessibility and mean-reversion. Only the Black non-Hispanic displacement model, examined conditional on baseline composition, recovers the elevation signal. The GWR then reveals that even this elevation–displacement relationship is geographically structured in ways the global model obscures, concentrating in northern communities rather than along the ridge. This reframes the within-ridge analysis (Section 3.4) not as a replication but as a distinct investigation of a

different spatial mechanism. Finally, the within-ridge OLS and GWR identify the interaction of historical vulnerability and elevation as most strongly associated with gentrification intensity at the sub-regional scale. This finding is invisible at the county level, where compositional dynamics in northern MDC dominate the spatial pattern. The implication for future climate gentrification research is that studies operating at a single geographic scale risk either missing the phenomenon entirely or misattributing it to the wrong communities and mechanisms.

Limitations

Several limitations should be noted. First, ACS 5-year estimates carry margins of error, particularly for small tract populations, and the rolling average structure introduces temporal smoothing that may obscure year-to-year dynamics within the 2010–2023 observation window. The comparison of 2010 ACS data (anchored to 2010 tract boundaries) with 2023 ACS data (anchored to 2020 tract boundaries) required population-weighted geographic harmonization to account for tract splits and mergers. Second, results are subject to the modifiable areal unit problem (MAUP), in which estimates can vary with the scale and zoning of spatial units [48], meaning the tract-level OLS and GWR results are specific to the 2020 census tract geometry. Similarly, tract-level mean elevation and a single inundation scenario do not capture within-tract or parcel-level exposure heterogeneity. Third, the outcome variables measure aggregate ecological change rather than individual household trajectories: tract-level income change cannot distinguish in-place income gains among incumbent residents from residential turnover, and the Black non-Hispanic share indicator does not capture the displacement of Black Hispanic households. This demographic indicator is used because segregation-era policy historically concentrated Black residents on the ridge—not because Black households are assumed to respond differently to market forces.

Additional limitations concern causal identification and generalizability. Including the 2010 baseline Black non-Hispanic share as a compositional control in the county-wide displacement model is methodologically necessary to account for ceiling effects on percentage-point change; however, because elevation and historical Black settlement overlap spatially, this control partially absorbs the baseline correlation between elevation and demographic geography. Consequently, the county-wide elevation association cannot be fully separated from historical settlement patterns and is reported conditional on baseline composition (Table A3). Furthermore, the cross-sectional change design cannot establish causal direction or isolate climate revaluation from co-occurring housing, zoning, insurance and resilience policies [11,13,25] nor can it rule out voluntary out-migration unrelated to housing costs or test whether climate gentrification amplifies earlier exclusionary policies. Finally, because these findings reflect Miami-Dade County's specific combination of a limestone ridge, porous subsurface hydrology and segregation history, applying this framework to other coastal regions requires a local recalibration of both physical elevation thresholds and socioeconomic vulnerability criteria.

5. Summary and Future Work

This study empirically examined the spatial dynamics of climate gentrification in Miami-Dade County, Florida, during 2010–2023, integrating demographic and environmental data across 646 census tracts to test whether SLR risk exposure is associated with differential income growth, home value appreciation and demographic displacement. The analysis proceeded through a nested four-stage framework: a county-wide descriptive comparison of Vulnerable Ridge Communities (VRCs), a continuous elevation gradient analysis, a county-wide OLS and GWR framework and a targeted within-ridge regres-

sion across the 277 high-elevation tracts of the Miami Rock Ridge. The main findings are as follows:

- In the county-wide regressions, demographic displacement was a more sensitive indicator of climate gentrification than unconditioned income change: mean tract elevation showed no association with aggregate county-wide income growth, where affluent ridge enclaves and baseline mean-reversion mask localized gains, but was significantly associated with Black non-Hispanic population decline where baseline Black non-Hispanic shares were high (-6.6 percentage points per meter in tracts at least 30% Black non-Hispanic in 2010; $p_{\mathrm{HC3}} = 0.001$)
- The county-wide GWR reveals pronounced spatial non-stationarity, with negative local elevation coefficients concentrating in majority-Black northern suburban communities rather than along the narrow Miami Rock Ridge, demonstrating how unstratified local kernels can reflect broader suburban compositional gradients while blurring narrow physical geographic boundaries.
- Vulnerable Ridge Communities (VRCs; $n = 42$) experienced 81.7% median income growth, 85.9% median home value appreciation and a 15.0 percentage-point decline in Black non-Hispanic population share, significantly outpacing both lower-elevation tracts and other high-elevation ridge tracts in MDC.
- Within the high-elevation corridor ($n = 277$), baseline socioeconomic vulnerability is a significant positive predictor of gentrification intensity across all ridge tracts; net of baseline vulnerability, mean elevation retains an independent positive association with compounding income growth and racial turnover ($\beta = 0.423$, $p = 0.007$, $p_{\mathrm{HC3}} = 0.001$) even among tracts that already exceed the 2.1 m (7 ft) planning threshold.
- A single-scale analytical approach would either miss the climate gentrification signal entirely (county income OLS) or misattribute it geographically (county GWR); the nested four-stage framework is necessary to recover the full spatial structure of the phenomenon.

Several directions for future research follow from these findings. The cross-sectional design documents cumulative gentrification outcomes over a 13-year period but cannot resolve year-by-year displacement and investment dynamics. Ongoing work incorporating residential building permit records for Miami-Dade County will enable a longitudinal tracking of physical redevelopment activity and allow for a direct assessment of whether construction investment precedes detectable demographic turnover, providing a leading indicator of neighborhood transition ahead of lagging ACS estimates. Property transaction and rental price records, together with household-level mobility data, would allow for a direct measurement of reinvestment and household movement rather than the compositional proxy provided by tract-level census estimates. Finally, the analytical framework developed here is transferable to other low-lying coastal metropolitan areas with comparable topographic and demographic profiles, where the intersection of historical segregation and SLR-driven elevation revaluation may be generating similar but as yet undocumented displacement dynamics. These findings relate to the United Nations Sustainable Development Goals (SDGs) [49], particularly SDG 10 (reduced inequalities), SDG 11 (sustainable cities and communities) and SDG 13 (climate action), as adaptation that concentrates investment on high ground can shift risk and displacement onto historically vulnerable residents.

Author Contributions: Conceptualization, Patrick Davenport; methodology, Patrick Davenport and Levente Juhász; software, Patrick Davenport; formal analysis, Patrick Davenport; data curation, Patrick Davenport; writing—original draft preparation, Patrick Davenport and Levente Juhász; writing—review and editing, Patrick Davenport and Levente Juhász; visualization, Patrick Davenport;

supervision, Levente Juhász. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Data and analysis scripts supporting the reported results are available from the corresponding author upon reasonable request.

Conflicts of Interest: L.J. is a member of the Editorial Board of *lISPRS International Journal of Geo-Information*. He did not participate in the peer review or editorial decision-making process for this manuscript in any capacity.

Abbreviations

The following abbreviations are used in this manuscript:

ACS	American Community Survey
AICc	Corrected Akaike Information Criterion
CBD	Central Business District
DEM	Digital Elevation Model
FEMA	Federal Emergency Management Agency
GI	Gentrification Intensity Index
GWR	Geographically Weighted Regression
HC3	Heteroscedasticity-Consistent Standard Errors (type 3)
MDC	Miami-Dade County
MGWR	Multiscale Geographically Weighted Regression
NAVD88	North American Vertical Datum of 1988
NHGIS	National Historical Geographic Information System
NOAA	National Oceanic and Atmospheric Administration
OLS	Ordinary Least Squares
SLR	Sea Level Rise
VIF	Variance Inflation Factor
VRC	Vulnerable Ridge Community
VS	Vulnerability Score

Appendix A. Robustness and Diagnostic Analyses

This appendix reports regression diagnostics, alternative spatial specifications and sensitivity analyses for the models in Section 3.3 (county-wide) and Section 3.4. Moran's I uses queen contiguity weights for county-wide models and k-nearest-neighbor weights ($k = 5$, row-standardized) for within-ridge models with 999 permutations. GWR significance uses the corrected critical t-value for multiple testing.

Table A1. OLS diagnostics for the county-wide and within-ridge models. Jarque–Bera and Shapiro–Wilk test the null of normally distributed residuals; Breusch–Pagan and White test the null of constant residual variance; for these tests, $p < 0.05$ indicates a violation. VIF screens for multicollinearity (values above 5 to 10 are problematic). Moran's I measures residual spatial autocorrelation (positive values indicate clustering of similar residuals), and the LM tests assess spatial lag and spatial error dependence.

	County: Income	**County: Black NH**	**Ridge: GI**
n	646	646	277
R^2/Adj. R^2	0.175/0.170	0.351/0.346	0.328/0.318
AIC/AICc	7029.2/7031.4	4455.3/4457.4	894.1/896.4

Table A1. *Cont.*

	County: Income	County: Black NH	Ridge: GI
Normality of residuals			
Jarque–Bera p	< 0.001	< 0.001	< 0.001
Shapiro–Wilk p	< 0.001	< 0.001	0.004
Skewness/Kurtosis	0.97/4.14	0.10/8.22	0.33/4.12
Heteroscedasticity			
Breusch–Pagan p	0.005	< 0.001	< 0.001
White p	< 0.001	< 0.001	< 0.001
Multicollinearity			
Maximum VIF	2.48	2.50	1.38
Spatial dependence			
Residual Moran's I (p)	0.137 (0.001)	0.168 (0.001)	0.205 (0.001)
LM-lag/Robust LM-lag p	0.004/< 0.001	0.004/< 0.001	0.002/< 0.001
LM-error/Robust LM-error p	< 0.001/< 0.001	< 0.001/< 0.001	< 0.001/< 0.001

Table A2. Comparison of global and local spatial specifications. AIC (OLS, spatial lag, spatial error) and AICc (OLS, GWR) are separate scales and are compared within each. MGWR was estimated on standardized variables and is reported for bandwidths only.

Model	County: Black NH	Ridge: GI
OLS	AIC = 4455.3; AICc = 4457.4 $R^2 = 0.351$; residual $I = 0.168$	AIC = 894.1; AICc = 896.4 $R^2 = 0.328$; residual $I = 0.205$
Spatial lag	AIC = 4450.8; $\rho = 0.129$ ($p = 0.022$)	AIC = 888.6; $\rho = 0.209$ ($p = 0.009$)
Spatial error	AIC = 4409.0; $\lambda = 0.432$ ($p < 0.001$)	AIC = 864.5; $\lambda = 0.483$ ($p < 0.001$)
GWR	AICc = 4301.9; bandwidth = 152 mean local $R^2 = 0.565$ (0.186–0.822) residual $I = -0.013$	AICc = 844.8; bandwidth = 96 mean local $R^2 = 0.487$ (0.241–0.669) residual $I = 0.036$
MGWR bandwidths	—	Intercept 55; Pct. Inundated 260; Elevation 276; Distance 276; VS 96

Table A3. Elevation association conditional on baseline Black non-Hispanic share (HC3 standard errors). Panel A: elevation × baseline share interaction (both centered) and implied elevation slopes. Panel B: models estimated within tracts above baseline share thresholds, using percentage-point change and change as a percentage of the baseline share.

Panel A: Interaction model ($n = 646$)		
Term	β	$p_{\mathbf{HC3}}$
Elevation (at mean baseline share)	−1.271	0.002
Baseline Black NH share	−0.185	< 0.001
Elevation × baseline share	−0.071	< 0.001
Elevation slope at baseline share	**pp per m**	p
25th percentile (0.2%)	−0.122	0.804
50th percentile (3.3%)	−0.338	0.467
75th percentile (16.1%)	−1.248	0.003
90th percentile (66.9%)	−4.860	< 0.001
Panel B: Historically Black tracts		
Baseline share threshold	**pp change per m**	**% of baseline per m**
$\geq 15\%$ ($n = 174$)	−5.932 ($p < 0.001$)	−8.015 ($p = 0.017$)
$\geq 30\%$ ($n = 128$), VRC criterion	−6.577 ($p = 0.001$)	−8.819 ($p = 0.020$)
$\geq 50\%$ ($n = 100$)	−7.298 ($p = 0.068$)	−10.220 ($p = 0.200$)

Table A4. County-wide GWR bandwidth sensitivity (Black non-Hispanic change). The AICc-optimal bandwidth is 152; 646 is the global model. Significant tracts use the corrected critical t-value (2.76 at the optimal bandwidth).

Bandwidth	AICc	Mean Local R^2	Residual I	Elev. Sig. Tracts	Of Which Negative	Mean Elev. β
60	4347.7	0.548	−0.076	57	57	−1.871
95	4315.2	0.549	−0.044	70	70	−1.467
131	4302.8	0.562	−0.023	98	98	−1.314
152	**4301.9**	**0.565**	**−0.013**	**120**	**120**	**−1.298**
200	4304.8	0.564	0.009	144	144	−1.211
300	4313.4	0.560	0.049	179	179	−1.113
646	4422.9	0.416	0.151	426	426	−1.184

Note: Bold indicates the AICc-optimal bandwidth.

At the optimal bandwidth, local multicollinearity was limited (maximum local VIF = 9.08; condition number above 30 in 13 of 646 tracts, standardized predictors). Excluding the baseline Black non-Hispanic control, 135 tracts had significant elevation coefficients, which were all negative.

Table A5. Sensitivity of the within-ridge model (Table 4) to the elevation cutoff. VS is restandardized within each sample; p-values use HC3 standard errors.

Cutoff (m)	Cutoff (ft)	n	Adj. R^2	Elev. β	Elev. p	VS β	VS p	Pct. Inund. p
1.50	4.9	539	0.281	0.345	0.004	0.459	< 0.001	0.198
1.80	5.9	423	0.301	0.411	0.001	0.452	< 0.001	0.137
2.13	**7.0**	**277**	**0.318**	**0.423**	**0.001**	**0.470**	$<$ **0.001**	**0.134**
2.40	7.9	179	0.348	0.414	0.005	0.548	< 0.001	0.740
2.70	8.9	127	0.393	0.373	0.028	0.618	< 0.001	0.956
3.00	9.8	68	0.513	0.439	0.223	0.816	< 0.001	0.610

Note: Bold indicates the cutoff used in the main analysis (2.13 m; 7 ft).

Table A6. Sensitivity of the within-ridge model to GI construction ($n = 277$). Components are standardized county-wide. Coefficients are divided by the standard deviation of each dependent variable; p-values use HC3 standard errors. Income change and Black non-Hispanic change are nearly uncorrelated ($r = 0.05$); the first principal component explains 52.6% of their variance with equal loadings (0.707), matching equal weights.

Index Variant	Spearman ρ with GI	Adj. R^2	Elev. β (Std.)	Elev. p	VS β (Std.)	VS p
GI (equal weights)	**1.000**	**0.318**	**0.290**	**0.001**	**0.322**	$<$ **0.001**
Income change only	0.718	0.074	0.054	0.850	0.095	0.005
Black NH change only	0.573	0.323	0.331	0.133	0.337	< 0.001
Black NH weight 0.5	0.966	0.226	0.217	0.159	0.257	< 0.001
Black NH weight 2	0.965	0.357	0.328	0.006	0.350	< 0.001
Rank-based index	0.926	0.200	0.004	0.976	0.273	< 0.001

Note: Bold indicates the index used in the main analysis (equal weights).

Table A7. Sensitivity of VRC delineation to income and Black non-Hispanic thresholds. Each row compares VRC tracts with the remaining ridge tracts (Mann–Whitney U). Black non-Hispanic change is not tested because it is conditioned by the selection criterion. Bold: manuscript definition.

Income Below	Black NH Above	n	Baseline Black NH (Median %)	Income Growth (%)	Income p	Home Value Apprec. (%)	Home Value p	Black NH Change (pp)
$35,000	25%	38	71.5	89.7	0.004	86.8	< 0.001	−16.6
$35,000	30%	35	73.1	92.9	0.004	86.8	< 0.001	−17.2
$35,000	35%	33	74.8	92.9	0.004	86.8	< 0.001	−17.4

Table A7. *Cont.*

Income Below	Black NH Above	*n*	Baseline Black NH (Median %)	Income Growth (%)	Income *p*	Home Value Apprec. (%)	Home Value *p*	Black NH Change (pp)
$40,000	25%	46	70.7	78.8	0.035	85.9	< 0.001	−14.7
$40,000	**30%**	**42**	**71.8**	**81.7**	**0.031**	**85.9**	$<$ **0.001**	**−15.0**
$40,000	35%	40	72.9	81.7	0.030	85.2	< 0.001	−15.7
$45,000	25%	56	71.0	68.2	0.257	85.9	< 0.001	−15.1
$45,000	30%	52	72.9	72.7	0.257	85.9	< 0.001	−15.6
$45,000	35%	50	73.9	72.7	0.263	85.2	< 0.001	−15.9

References

1. Nerem, R.S.; Beckley, B.D.; Fasullo, J.T.; Hamlington, B.D.; Masters, D.; Mitchum, G.T. Climate-change–driven accelerated sea-level rise detected in the altimeter era. *Proc. Natl. Acad. Sci. USA* **2018**, *115*, 2022–2025. https://doi.org/10.1073/pnas.1717312115.
2. Parkinson, R.W.; Wdowinski, S. A unified conceptual model of coastal response to accelerating sea level rise, Florida, U.S.A. *Sci. Total Environ.* **2023**, *892*, 164448. https://doi.org/10.1016/j.scitotenv.2023.164448.
3. Keenan, J.; Hill, T.; Gumber, A. Climate gentrification: From theory to empiricism in Miami-Dade County, Florida. *Environ. Res. Lett.* **2018**, *13*, 054001. https://doi.org/10.1088/1748-9326/aabb32.
4. Juhász, L.; Hochmair, H.H.; Santana, S.A.d.; Fu, Z.J. Sea Level Rise Impact Assessment Tool—A Web-Based Application for Community Resilience in Coral Gables, Florida. *Int. J. Spat. Data Infrastructures Res.* **2020**. *15*, 36–55. https://doi.org/10.2902/1725-0463.2020.15.art2.
5. Lee, S.; Lin, J. Natural Amenities, Neighbourhood Dynamics, and Persistence in the Spatial Distribution of Income. *Rev. Econ. Stud.* **2018**, *85*, 663–694. https://doi.org/10.1093/restud/rdx018.
6. Mohl, R.A. Whitening Miami: Race, Housing, and Government Policy in Twentieth-Century Dade County. *Fla. Hist. Q.* **2001**, *79*, 319–345.
7. NOAA Office for Coastal Management. Sea Level Rise Data: 1–10 ft Sea Level Rise Inundation Extent. InPort. 2025. Available online: https://www.fisheries.noaa.gov/inport/item/48106 (accessed on 28 July 2026).
8. George, P.S. Colored Town: Miami's Black Community, 1896–1930. *Fla. Hist. Q.* **1977**, *56*, 4.
9. Bin, O.; Kruse, J.B. Real Estate Market Response to Coastal Flood Hazards. *Nat. Hazards Rev.* **2006**, *7*, 137–144. https://doi.org/10.1061/(ASCE)1527-6988(2006)7:4(137).
10. Kousky, C. Learning from Extreme Events: Risk Perceptions after the Flood. *Land Econ.* **2010**, *86*, 395–422. https://doi.org/10.3368/le.86.3.395.
11. Gourevitch, J.D.; Kousky, C.; Liao, Y.P.; Nolte, C.; Pollack, A.B.; Porter, J.R.; Weill, J.A. Unpriced climate risk and the potential consequences of overvaluation in US housing markets. *Nat. Clim. Change* **2023**, *13*, 250–257. https://doi.org/10.1038/s41558-023-01594-8.
12. U.S. Census Bureau. *American Community Survey 5-Year Estimates, Table B25003: Tenure, 2023*; 2006–2010 and 2019–2023 releases, retrieved via the Census Data API; U.S. Census Bureau: Suitland, MD, USA, 2023.
13. Taylor, Z.J.; Aalbers, M.B. Climate Gentrification: Risk, Rent, and Restructuring in Greater Miami. *Ann. Am. Assoc. Geogr.* **2022**, *112*, 1685–1701. https://doi.org/10.1080/24694452.2021.2000358.
14. Aune, K.T.; Gesch, D.; Smith, G.S. A spatial analysis of climate gentrification in Orleans Parish, Louisiana post-Hurricane Katrina. *Environ. Res.* **2020**, *185*, 109384. https://doi.org/10.1016/j.envres.2020.109384.
15. Best, K.; Jouzi, Z. Climate Gentrification: Methods, Gaps, and Framework for Future Research. *Front. Clim.* **2022**, *4*,828067 . https://doi.org/10.3389/fclim.2022.828067.
16. Freeman, L. Displacement or Succession?: Residential Mobility in Gentrifying Neighborhoods. *Urban Aff. Rev.* **2005**, *40*, 463–491. https://doi.org/10.1177/1078087404273341.
17. Best, K.B.; Jouzi, Z.; Islam, M.S.; Kirby, T.; Nixon, R.; Hossan, A.; Nyiawung, R.A. Typologies of multiple vulnerabilities and climate gentrification across the East Coast of the United States. *Urban Clim.* **2023**, *48*, 101430. https://doi.org/10.1016/j.uclim.2023.101430.
18. de Koning, K.; Filatova, T. Repetitive floods intensify outmigration and climate gentrification in coastal cities. *Environ. Res. Lett.* **2020**, *15*, 034008. https://doi.org/10.1088/1748-9326/ab6668.
19. See, J.; Cuaton, G.P.; Wilmsen, B.; Peja, P.J. Uncovering the drivers of climate gentrification in the Global South: Case study of Tacloban City, Philippines. *Political Geogr.* **2025**, *117*, 103275. https://doi.org/10.1016/j.polgeo.2025.103275.
20. Blok, A. Urban green gentrification in an unequal world of climate change. *Urban Stud.* **2020**, *57*, 2803–2816. https://doi.org/10.1177/0042098019891050.

21. Shokry, G.; Anguelovski, I.; Connolly, J.J.T.; Maroko, A.; Pearsall, H. "They Didn't See It Coming": Green Resilience Planning and Vulnerability to Future Climate Gentrification. *Hous. Policy Debate* **2022**, *32*, 211–245. https://doi.org/10.1080/10511482.2021.1944269.
22. Seeteram, N.A.; Ash, K.; Sanders, B.F.; Schubert, J.E.; Mach, K.J. Modes of climate mobility under sea-level rise. *Environ. Res. Lett.* **2023**, *18*, 114015. https://doi.org/10.1088/1748-9326/acfe22.
23. Southeast Florida Regional Climate Change Compact. 2024 Statement of Continued Use of the 2019 Southeast Florida Regionally Unified Sea Level Rise Projection. Technical Report. 2024. Available online: https://southeastfloridaclimatecompact.org/wp-content/uploads/2024/12/2024-SLR-Statement_120924_FINAL.pdf (accessed on 17 July 2026).
24. Greater Miami & The Beaches. Resilient 305 Strategy. Technical Report, Miami-Dade. 2019. Available online: https://www.miamidade.gov/global/environment/resilience/resilient305.page (accessed on 2 August 2026).
25. Seeteram, N.A.; Shi, L.; Mach, K.J.; Carrère, A.; Chatterjee, T.; Garner, A.; Horton, R.M. Challenges and opportunities in scaling climate-resilient housing solutions in the United States. *Nat. Commun.* **2026**, *17*, 2032. https://doi.org/10.1038/s41467-026-68595-x.
26. University of Miami Libraries. Overtown Chronology. Oral History Project. Available online:
27. University of Miami. Housing Timeline: Race and Property; Changing Neighborhoods. Available online: https://affordablehousing.miami.edu/housing-timeline/race-property/index.html (accessed on 14 March 2026).
28. Yarveysi, F.; Jafarzadegan, K.; Tripathy, S.S.; Moftakhari, H.; Moradkhani, H. A data-driven framework for an efficient block-level coastal flood risk assessment. *Int. J. Disaster Risk Reduct.* **2025**, *122*, 105478. https://doi.org/10.1016/j.ijdrr.2025.105478.
29. Scanniello, N. *New American Community Survey 5-Year Estimates for the First Time Provide Data on Three Time Periods That Don't Overlap*; U.S. Census Bureau: Suitland, MD, USA, 2020.
30. Cahill, M.; Franklin, R.S. Homeownership, Home Foreclosure, and Nativity: Evidence from Miami-Dade County, Florida. *J. Reg. Sci.* **2013**, *53*, 91–117. https://doi.org/10.1111/jors.12014.
31. Haslag, P.; Weagley, D. From L.A. to Boise: How Migration Has Changed During the COVID-19 Pandemic. *J. Finance Quant. Anal.* **2024**, *59*, 2068–2098. https://doi.org/10.1017/S002210902300073X.
32. Ozgen, E.; Murzyn, Z. The Impact of COVID-19 on the Real Estate Market in Florida: Insights from Opportunity Recognition Theory. *J. Business, Ind. Econ.* **2024**, *29*, 4.
33. Juhász, L.; Xu, J.; Parkinson, R.W. Beyond the Tide: A Comprehensive Guide to Sea-Level-Rise Inundation Mapping Using FOSS4G. *Geomatics* **2023**, *3*, 522–540. https://doi.org/10.3390/geomatics3040028.
34. NOAA. *Technical Considerations for Use of Geospatial Data in Sea Level Change Mapping and Assessment*; NOAA Technical Report NOS 2010-1; NOAA: Washington, DC, USA, 2010.
35. Miami-Dade County Open Data Hub. 2021 Miami-Dade County DEM 5ft—Overview. 2021. Available online: https://mdc.maps.arcgis.com/home/item.html?id=8c48d4bb8d9a42908f4936f698a2961a#overview (accessed on 13 December 2025).
36. Schroeder, J.P. Target-Density Weighting Interpolation and Uncertainty Evaluation for Temporal Analysis of Census Data. *Geogr. Anal.* **2007**, *39*, 311–335. https://doi.org/10.1111/j.1538-4632.2007.00706.x.
37. Logan, J.R.; Xu, Z.; Stults, B.J. Interpolating U.S. Decennial Census Tract Data from as Early as 1970 to 2010: A Longitudinal Tract Database. *Prof. Geogr.* **2014**, *66*, 412–420. https://doi.org/10.1080/00330124.2014.905156.
38. Mennis, J. Generating Surface Models of Population Using Dasymetric Mapping. *Prof. Geogr.* **2003**, *55*, 31–42. https://doi.org/10.1111/0033-0124.10042.
39. Compact, S.F.R.C.C. Unified Sea Level Rise Projection—Southeast Florida. Technical Report. 2020. Available online: https://southeastfloridaclimatecompact.org/wp-content/uploads/2023/10/2019-sea-level-projections.pdf (accessed on 17 July 2026).
40. Sweet, W.V.; Hamlington, R.; Kopp, C.; Weaver, P. *Global and Regional Sea Level Rise Scenarios for the United States: Updated Mean Projections and Extreme Water Level Probabilities Along U.S. Coastlines*; NOAA Technical Report NOS 01; National Oceanic and Atmospheric Administration, National Ocean Service: Silver Spring, MD, USA, 2022.
41. Beck, K. Gentrification and Neighborhood Housing Wealth: How Gentrification Reproduces the Racial Stratification of Urban Neighborhoods. *Socius* **2024**, *10*, 1–16. https://doi.org/10.1177/23780231241234645.
42. Kim, S.K.; Park, S. How does exposure to climate risk contribute to gentrification? *Cities* **2023**, *137*, 104321. https://doi.org/10.1016/j.cities.2023.104321.
43. Hwang, J.; Sampson, R.J. Divergent Pathways of Gentrification: Racial Inequality and the Social Order of Renewal in Chicago Neighborhoods. *Am. Sociol. Rev.* **2014**, *79*, 726–751. https://doi.org/10.1177/0003122414535774.
44. Long, J.S.; Ervin, L.H. Using Heteroscedasticity Consistent Standard Errors in the Linear Regression Model. *Am. Stat.* **2000**, *54*, 217–224. https://doi.org/10.1080/00031305.2000.10474549.
45. Brunsdon, C.; Fotheringham, A.S.; Charlton, M.E. Geographically Weighted Regression: A Method for Exploring Spatial Nonstationarity. *Geogr. Anal.* **1996**, *28*, 281–298. https://doi.org/10.1111/j.1538-4632.1996.tb00936.x.

46. Oshan, T.M.; Li, Z.; Kang, W.; Wolf, L.J.; Fotheringham, A.S. mgwr: A Python Implementation of Multiscale Geographically Weighted Regression for Investigating Process Spatial Heterogeneity and Scale. *ISPRS Int. J. Geo-Inf.* **2019**, *8*, 269. https://doi.org/10.3390/ijgi8060269.
47. Ley, D. Alternative Explanations for Inner-City Gentrification: A Canadian Assessment. *Ann. Assoc. Am. Geogr.* **1986**, *76*, 521–535. https://doi.org/10.1111/j.1467-8306.1986.tb00134.x.
48. Openshaw, S. *The Modifiable Areal Unit Problem*; Number 38 in Concepts and Techniques in Modern Geography, Geo Books: Norwich, UK, 1984.
49. United Nations. *Transforming Our World: The 2030 Agenda for Sustainable Development*; Resolution A/RES/70/1; United Nations General Assembly: New York, NY, USA, 2015.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.